Visual Attention & Body Image Disturbance: A Small-Scale Feasibility Test for Tracking Gaze During Body-Oriented Short-Form Video Content Consumption

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Abstract

Body image disturbance (BID) is associated with distinct visual attention patterns while viewing static images of people's bodies; however, it remains unclear whether this observation extends to dynamic media contexts, such as when watching body-oriented short-form video content. As a precursor to a planned between-subjects study comparing the gaze behavior of women with and without BID-driven conditions, we conducted a small-scale feasibility test of our proposed gaze-tracking methodology. Using a Tobii Eye Tracker 5 and a Dell S2421NX monitor, we recruited seven participants to complete directed (instructed fixation on six bodily areas of interest) and undirected (free-viewing) gaze tasks while watching body-oriented short-form videos that varied in body visibility and movement speed. We tested our hardware setup, a calibration validation step introduced partway through testing, and several potential gaze-analysis metrics, including raw gaze, fixation, and dwell-episode hit rates. A dwell-episode hit rate designed to capture sustained gaze on moving AOIs was better suited to dynamic targets than fixation or exploratory pursuit-based metrics, whose estimates were not sufficiently stable or interpretable in our data, given the ≈33 Hz gaze-point stream of our tracker, short runs, and annotation noise. In this small, non-randomized sample, participants who passed calibration validation had higher pooled raw-gaze hit rates than participants tested before validation was introduced (0.72 vs. 0.53), motivating a mandatory calibration check in the full study. Preliminary, hypothesis-generating patterns from the undirected task (n=2) suggest a potential face/head viewing bias, with most fixations falling outside the annotated body regions. We conclude that our pipeline is feasible for the planned full study, contingent on mandatory calibration validation, a revised AOI annotation approach, and continued use of dwell-episode hit rate.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; **Interaction techniques**.

Keywords

gaze tracking, social media, short-form video, body image



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1 Background

1.1 Body Image Disturbance & Visual Attention

Body image disturbance (BID) is a complex, multidimensional psychological phenomenon that expresses as the intense distortion of and preoccupation with one's own physical appearance [8]. Distinct from general aesthetic concern or bodily dissatisfaction, clinically-significant BID is structured around several overlapping dimensions, including cognitive and perceptual, that reinforce the phenomenon over time [1, 5]. Individuals who are acculturated to internalize observers' perspectives of their bodies (a process termed "self-objectification"), notably women and girls, are at increased risk of developing BID [3, 17]. Self-objectification is realized through "habitual body monitoring" and "body surveillance," in which appearance is continually monitored and evaluated against internalized ideals [3]. Sustained by physical comparison [17], this surveillance reinforces dissatisfaction and can lead to compulsive body-checking of one's own body and the bodies of others, as seen in eating disorders for which BID is a clinically-significant feature [19].

Previous work suggests that women with BID-driven conditions, such as anorexia nervosa and bulimia nervosa, exhibit distinct gaze fixation patterns when viewing images of their own and others' bodies, including increased visual attention to "weight-indicative" areas (e.g., hips, abdomen, thighs) and to areas they perceive as "unattractive" [4, 8–10, 14]. Early research into the connection between visual attention and BID even suggested that gaze tracking may someday be used in the assessment and monitoring of body shape dissatisfaction, given their demonstrated relationship [4].

1.2 Short-Form Video & BID

Short-form video platforms (SFVPs), such as TikTok and Instagram Reels, are social media platforms characterized by vertically oriented, short video content delivered through infinitely scrollable feeds. Their design encourages rapid consumption of videos in succession, or "binge-scrolling," where users passively view large numbers of recommended videos in a single feed for extended periods without seeking out specific content [21]. These platforms are

widely used by adolescents and young adults [6], including those in eating-disorder treatment, of which many indicate that SFVPs are their primary or favored social media platforms [18]. SFVPs are saturated with person- and body-centric content. This includes intentionally body-focused posts like workout videos, makeup tutorials, and weight-loss transformations, as well as unintentionally body-centric content such as casual outfit display posts or "get ready with me" (GRWM) routine videos. This constant exposure to body-oriented material has been speculated to impact individuals susceptible to BID. A growing number of papers associate short-form video exposure to bodily dissatisfaction, appearance comparison, and negative mood in nonclinical samples [7, 11]; however, findings remain mixed, and the effects on clinically susceptible populations (as well as the attentional mechanisms underlying them) remain poorly understood. Additionally, distinct from easily-recorded and traditionally-studied social media interactions, including "likes," "comments," and "shares," SFVP use primarily involves passive visual attention during extended scroll periods. This gaze-based mode of engagement, though prevalent, is largely unexplored despite established links between BID, visual attention, and SFVP use in susceptible populations.

2 Goals & Objectives

To understand whether gaze behaviors observed in static contexts translate to dynamic short-form video, we plan to conduct a between-subjects study comparing the gaze patterns of women with and without BID-driven conditions while they watch body-oriented SFVs. If reliable group differences are observed relative to validated BID measures, the findings could inform investigation and development of adaptive in-feed interventions. As a precursor, we conducted this small-scale feasibility test with a convenience sample of young women (not screened for BID), evaluating our hardware, gaze-task design, and prospective metrics for quantifying gaze on bodily areas of interest (AOIs) in SFV content varying in body visibility and movement speed.

To our knowledge, the study of gaze on bodily AOIs in BID and BID-driven eating disorders has focused almost exclusively on gaze upon static images [14], virtual reality/avatar-based environments [2], or real-world interactions via mirror exposure [22], as opposed to video-based content, such as body-oriented short-form video. Because established metrics for quantifying gaze on bodily AOIs were developed for static stimuli, it is unclear whether they would be able to transfer to dynamic AOIs that move across the frame. Therefore, evaluating prospective metrics for this setting across body-visibility and movement-speed conditions was a central aim of this feasibility test, alongside testing our overall procedural methodology. The present test addresses this measurement prerequisite, while the planned study will use the refined pipeline to test whether previously documented BID-related differences in attention to bodily regions extend to dynamic SFVs.

3 Methods

3.1 Video Feed & Annotation

We sourced videos used in our feed from the SnapUGC dataset, a publicly available collection of user-generated SFVs from Snapchat Spotlight [16]. We selected videos featuring a single female-perceived person as the subject of the video without on-screen text to minimize visual distractions. To resemble the variety of body-oriented content encountered on SFVPs, we included videos across a variety of body visibility (full-body, upper-body), movement speed (little/none, slow/infrequent, fast/frequent), and skin visibility (little/none, some, much). We initially screened videos using a combination of YOLOv8 object detection and EasyOCR for text recognition, allowing us to rapidly identify potential candidates before manual screening. We further categorized videos by the visibility of six AOIs. Five (clavicles/collarbones, upper arms, stomach/abdomen/waist, hips, and thighs) were informed by prior static-image gaze-tracking research regarding BID [9]. We additionally included the jawline/neck, reflecting the growing interest toward the jawline in contemporary online appearance culture (e.g., the "looksmaxxing" trend [24]). To ensure consistent viewing conditions, we assembled videos into an auto-scrolling feed where each video played for five seconds before advancing to the next. In the vein of minimizing visual distractions, we omitted on-screen elements such as "like," "comment," and "share" buttons from the feed.

We annotated bodily AOIs within videos using Label Studio (version 1.16.0) by drawing bounding boxes around AOIs at key frames. To create continuous annotations for each frame, we exported the key frame annotations and linearly interpolated the bounding box coordinates across all intermediary frames. This resulted in per-frame bounding box coordinates, which we then used to analyze gaze point containment within each AOI bounding box. Bounding boxes were drawn at moments of movement and around rapid cuts or scroll transitions. Nevertheless, because interpolation was not formally segmented at every transition, we cannot exclude the possibility that some interpolated boxes crossed frames in which the corresponding AOI was absent or occupied intervening space.

3.2 Participants

We recruited a convenience sample of seven (7) female participants to take part in this feasibility test. All participants were aged 30 or younger (mean: 25, min: 20, max: 30), and most (71%) scrolled on a SFVP several times a day. Two had blue/green eyes (29%), while the remaining participants had brown eyes.

3.3 Gaze Calibration, Validation & Session Flow

We recorded gaze at ≈33 Hz using a Tobii Eye Tracker 5, mounted beneath a Dell S2421NX monitor via a custom 3D-printed mount, with participants seated ≈25 inches away. After standard calibration via the Tobii Experience desktop application, the final three participants additionally completed a 7-point calibration validation task (Pass: mean error $\leq 1.0^\circ$; Warn: $\leq 2.0^\circ$; Fail: $>2.0^\circ$, requiring recalibration), introduced mid-study to address gaze-accuracy variability we observed in preliminary testing. Because validation ran before each task, a non-passing result could be addressed by recalibrating and re-validating, making calibration quality an in-session correction rather than a post-hoc diagnostic. In this test, a "Fail" prompted recalibration, whereas a "Warn" was allowed to proceed.

Each session included an undirected (free-viewing) gaze task and a directed gaze task, in which participants were instructed (on-screen and verbally) to fixate on specific AOIs. Calibration

and validation were administered at the start of each gaze task, so validation outcomes were session-specific rather than fixed per participant.

We chose the Tobii Eye Tracker 5 for its in-lab availability and prior use in the literature [12]. Whether its modest ≈33 Hz gaze-point output could support interpretable measurement of gaze overlap on dynamic AOIs was itself part of this feasibility test.

3.4 Gaze-Based Hit Rate Metrics

We adopted three complementary hit-rate metrics: (1) raw gaze, (2) fixation, and (3) dwell-episode.

Raw gaze hit rate was the proportion of gaze samples whose coordinates fell within an AOI's annotated bounding box, among all samples while that AOI was present and annotated. For annotations where two bounding boxes existed in the same frame for the same AOI (e.g., two "upper arm" bounding boxes, each representing a visible arm), a hit in either box was counted as a hit for that AOI. This is a classical AOI-relevant metric that may be adaptable to the dynamic format; however, it is sensitive to saccadic noise.

Fixation hit rate applied the same containment logic to fixations identified via a Dispersion-Threshold Identification (I-DT) algorithm, where for each candidate window, dispersion was calculated as $D = (\max x - \min x) + (\max y - \min y)$, and the window was extended until adding another valid gaze sample would exceed the dispersion threshold. We used a threshold of 40 px (≈0.44° visual angle) and a 100 ms minimum first-to-last-sample duration. Pixels were measured in the 1080×2424 stimulus-video coordinate space, in which approximately 90 video pixels corresponded to 1° of visual angle at the ≈25-inch viewing distance; the portrait video was scaled to ≈0.45× to fit screen height. Invalid gaze samples were excluded before fixation detection, and elapsed timestamps were used to calculate fixation duration. The implementation was based on Salvucci & Goldberg [20]; the 40 px threshold was specific to this study and yielded broadly similar descriptive patterns across 0.44°-1.33° in sensitivity analysis. A fixation was counted as a hit if its gaze-sequence centroid fell within the AOI's annotated bounding box on the video frame corresponding to fixation onset. I-DT is a classical and commonly used fixation-identification algorithm; however, because it identifies fixations based on low gaze-point dispersion in stationary screen coordinates, sustained gaze on a moving AOI may not be classified as a fixation when following the AOI produces screen-space dispersion beyond the predefined threshold.

Dwell-episode hit rate captured sustained gaze on a moving AOI that screen-space fixation detection may exclude. It was computed over episodes of in-box raw-gaze samples, allowing gaps of up to ≈100 ms between successive in-box samples and requiring at least 100 ms between the first and last in-box samples. It used the same denominator as raw-gaze hit rate, so dwell-episode hit rate ≤ raw-gaze hit rate by design. We additionally evaluated minimum-duration thresholds of 200 and 300 ms while retaining the ≈100 ms hit-gap tolerance.

3.5 Measuring Attention to Moving AOIs

As mentioned, screen-space fixation detection may underestimate sustained attention to or gaze overlap with a moving AOI, because gaze following a moving target can exhibit high screen-space dispersion and be discarded by I-DT as non-fixational. In our case, we were initially interested in whether some of this excluded behavior reflected "smooth pursuit," the continuous velocity-matched eye movement that stabilizes a moving target on the fovea.

Many smooth-pursuit classification methods use estimates of momentary eye velocity, including velocity-thresholded extensions of I-DT such as I-VDT [15]. In our data, the ≈33 Hz gaze-point stream provided limited temporal resolution for per-sample velocity estimation. We therefore explored three pursuit-oriented alternatives: two velocity-based pursuit metrics, direction-congruent velocity correlation (whether gaze moved in the AOI's direction) and pursuit gain (whether it matched the AOI's speed, measured as the ratio of gaze speed to AOI speed), as well as an AOI-relative fixation detector (whether gaze remained stable relative to the AOI). None provided a sufficiently stable or interpretable measure for inclusion in the primary analysis. First, direction-congruent velocity correlation was not clearly distinguishable from a circularly shifted null baseline. This could indicate that tracking was saccadic (as opposed to smooth), temporal resolution was limited, annotations were noisy, or a combination of these factors. Second, pursuit gain became unstable as targets slowed (with its gaze-to-target speed ratio diverging near zero). Finally, the AOI-relative detector injected the box's frame-to-frame annotation jitter (resulting in large part from rapid video cuts and scroll transitions) into the gaze trajectory, fragmenting fixations that were actually consistent on screen.

Importantly, these results point to a more basic observation: keeping gaze on a moving AOI may not require smooth pursuit. Because AOIs are extended regions rather than points, a viewer can sustain attention on a moving body part through a sequence of saccades that sample different parts of it, each landing inside the box, without ever matching the target's velocity. What matters for measuring sustained attention to an AOI is not the oculomotor mechanism, but whether gaze remained on the AOI over time. Dwell-episode hit rate asks only whether gaze stayed inside the AOI's bounding box for a sustained interval, with qualifying episodes lasting at least 100 ms, allowing gaps of up to ≈100 ms between successive in-box samples. This is a positional test that requires no velocity estimation, is less directly dependent on screen-space stability than I-DT, and needs no classification of gaze into fixations, saccades, or pursuit. As such, we adopted it as our positional measure of sustained gaze on moving AOIs. A tracker with a higher sampling rate, together with an annotation approach that better handles rapid cuts and scroll-induced motion, could make the velocity-based metrics viable in future work, if distinguishing smooth pursuit from saccadic tracking ever becomes a question of interest.

3.6 Directed vs. Undirected Analyses

For directed-gaze analyses, we stratified hit rates by AOI, movement speed, body visibility, and calibration-validation status, and compared fixation and dwell-episode hit rates by movement speed. Because participants were directed where to look, these analyses assessed the reliability of gaze-based AOI hit-rate measurement with the Tobii Eye Tracker 5 across movement speeds and calibration-validation outcomes. For undirected-gaze analyses (n=2, validation-pass participants only), we examined the same stratifications with

skin visibility, the proportion of fixations landing on vs. outside AOIs, and dwell retention (dwell-episode / raw hit rate) by movement speed. Because gaze was unconstrained, these analyses were descriptive and offered a first look at natural gaze behavior during body-oriented SFV viewing.

Given the feasibility focus and unequal eligible observation counts across participants and conditions, we report hit rates as pooled rates (total hits divided by total eligible observations) rather than averages of per-participant rates. Therefore, participants contribute proportionally to their eligible observation counts.

4 Results

4.1 Directed Gaze

Among the factors we examined, calibration-validation status was associated with the largest differences in hit rates. Validated-pass participants had higher raw gaze (0.715) and fixation (0.818) hit rates than unvalidated (0.525 / 0.587) or the single validated-warn (0.499 / 0.541) participant (Table 1). Because validation was introduced mid-study in a small, non-randomized sample, this is confounded with individual differences and recruitment order, so we interpret it as motivating a mandatory validation check rather than as evidence that validation itself raises accuracy. Among validated-pass participants, the median distance of missed points from the instructed AOI boundary was ≈1° of visual angle (0.95° for fixations and 1.03° for raw gaze), which may be interpreted as near-misses. However, we cannot meaningfully distinguish whether these near misses are the result of calibration or tracker error, annotation error, or failure to follow the directed-gaze instruction.

Slow/infrequent movement had the highest hit rates in every validation group. Among validated-pass participants, upper-body-framed stimuli yielded higher pooled hit rates than full-body-framed stimuli across all three metrics.

Because dwell-episode and raw-gaze hit rates share a denominator (all in-window gaze samples), their ratio (dwell retention) describes how much on-target gaze was sustained rather than momentary. Pooled across participants, dwell retention was 0.983 at the primary 100 ms minimum duration threshold (dwell-episode hit rate: 0.567 vs. raw-gaze hit rate: 0.576). Retention remained high across movement conditions and validation groups, ranging from 0.972 to 0.997. Because the 100 ms minimum duration and ≈100 ms hit-gap tolerance favor high retention, we repeated the directed-task analysis using stricter minimum durations while retaining the same gap tolerance. Pooled retention remained 0.963 at 200 ms and 0.942 at 300 ms. At 300 ms, retention was 0.918 for the no-validation group, 0.970 for the validated-pass group, and 0.960 for the validated-warn participant. Therefore, most on-target observations remained part of sustained episodes, even under stricter duration requirements. The difference between fixation and dwell-episode hit rates was also smaller during AOI motion (0.050-0.099) compared to little/no movement (0.087-0.144) across the three validation groups. Because fixation and dwell-episode hit rates have different denominators, we interpret this comparison qualitatively. Together with the positional rationale described in Section 3.5, these findings support using dwell-episode hit rate as a motion-robust complement to fixation for moving AOIs on SFVs in our full study.

4.2 Undirected Gaze (Hypothesis-Generating, n=2)

In our small, exploratory analysis of free-viewing gaze for the two participants who passed calibration validation, both showed their highest raw gaze hit rates on the jawline/neck, which was also highest for fixations pooled across participants (though P2's single highest fixation AOI was the stomach/abdomen/waist), followed by the stomach/abdomen/waist, then thighs (Table 2). However, most identified fixations fell outside any of the six annotated AOIs (P1: 70%, P2: 82%). For both participants, AOI hit rates were higher for full-body than upper-body content across raw gaze, fixations, and absolute fixation counts. One participant's raw gaze AOI hit rate increased with skin visibility (Little/none: 0.096 → Much: 0.117), while the other showed a modest net decrease (0.091 → 0.084).

Pooled across AOIs, dwell retention (the proportion of gaze-AOI overlap reflecting a sustained ≥100 ms dwell rather than a brief pass-through) was highest during fast/frequent movement for both participants (pooled ≈86% vs. ≈80% for the slower conditions). This fast-movement observation in dwell retention held at the AOI level for jawline/neck, upper arms, and stomach/abdomen/waist, but not for thighs, which showed the opposite pattern.

5 Discussion

5.1 Calibration Validation

Calibration-validation status was associated with larger differences in hit rates than any AOI, movement-speed, or body-visibility stratification. Adding the validation made calibration quality actionable as a non-passing result could be addressed on the spot by recalibrating and re-validating. Consistent with this, our one validated-warn participant, who proceeded without recalibrating, had among the lowest hit rates (including a 0.172 AOI cell). This motivates a stricter protocol for the full study in which we recalibrate until we achieve a "Pass," rather than proceed on a "Warn."

5.2 Movement Speed & Dwell Retention

Directed hit rates were highest for slow/infrequent movement, exceeding both little/none and fast/frequent. Given the sample size, we tentatively speculate that a slowly moving target may offer a continuous, low-demand anchor for gaze, whereas prolonged fixation on a static target is vulnerable to attentional and ocular drift [23] and faster motion is harder to follow. Additionally, the dwell-retention result suggests that the dwell-episode metric behaves as intended. On-target gaze was sustained rather than momentary, and remained so when AOIs moved. Because the 100-ms specification favors high retention, we conducted more informative sensitivity checks using 200 and 300 ms minimum durations, which yielded pooled retention of 0.963 and 0.942, respectively. Together with the conceptual distinction between positional AOI overlap and screen-space fixation, these results support using dwell-episode hit rate as a primary metric in the full study to describe sustained overlap with moving AOIs.

5.3 AOI Size

Because a larger bounding box is easier for a gaze estimate to fall within, hit-rates depend partly on AOI size, which varied in

Table 1: Pooled hit rate by stratification variable, metric, and calibration validation status (directed gaze task)

Stratification	Metric	No Validation (n=4)	Validated (Pass) (n=2)	Validated (Warn) (n=1)
Overall				
	Raw Gaze	0.525	0.715	0.499
All	Fixation	0.587	0.818	0.541
	dwell-episode	0.514	0.709	0.489
By AOI (body visibility)				
	Raw Gaze	0.499	0.838	0.737
Jawline/Neck (upper body)	Fixation	0.514	0.970	0.800
	dwell-episode	0.481	0.836	0.727
	Raw Gaze	0.538	0.673	0.407
Clavicles (upper body)	Fixation	0.569	0.778	0.506
	dwell-episode	0.529	0.666	0.397
	Raw Gaze	0.413	0.730	0.699
Upper Arms (upper body)	Fixation	0.511	0.807	0.867
	dwell-episode	0.401	0.724	0.685
	Raw Gaze	0.606	0.690	0.172
Stomach/Abdomen/Waist (full body)	Fixation	0.686	0.788	0.159
	dwell-episode	0.597	0.681	0.152
	Raw Gaze	0.452	0.662	0.384
Hips (full body)	Fixation	0.507	0.722	0.446
	dwell-episode	0.442	0.655	0.379
	Raw Gaze	0.638	0.699	0.598
Thighs (full body)	Fixation	0.740	0.841	0.545
	dwell-episode	0.628	0.695	0.597
By Movement Speed				
	Raw Gaze	0.485	0.594	0.508
Little/none	Fixation	0.582	0.728	0.582
	dwell-episode	0.474	0.584	0.495
	Raw Gaze	0.556	0.772	0.509
Slow/infrequent	Fixation	0.660	0.855	0.617
	dwell-episode	0.545	0.770	0.503
	Raw Gaze	0.537	0.697	0.500
Fast/frequent	Fixation	0.546	0.807	0.482
	dwell-episode	0.526	0.688	0.486

Note. Raw gaze and dwell-episode hit rates share a denominator (all in-window gaze samples), so dwell-episode ≤ raw gaze. Fixation hit rate is computed over detected fixations.

this feasibility test with both body region and body visibility (full vs. upper-body shots, i.e., body visibility). Differences in hit rate across AOIs therefore reflect box size in addition to attention, so we interpret them cautiously, and plan to normalize for or control bounding-box area and count in the full study.

5.4 Undirected Gaze

The following observations are drawn from the two participants who passed calibration validation (n=2) and are intended as hypothesis-generating rather than population-level claims (note that participants were not screened for BID). The prominence of the jawline/neck across raw gaze and fixation hit rates may reflect a general face/head bias in free viewing, possibly related to social-information-seeking tendencies [13]. The observation that most fixations fell outside any annotated AOI suggests AOI hit rate could be read as relative prioritization among a pre-specified set of regions as opposed to a proxy for overall visual interest in the body. In the full study, we will include a comparison baseline, such as a non-body control region or screen-center reference, to assess whether AOI overlap exceeds allocation expected from generic free-viewing biases. Higher AOI hit rates for full-body content may reflect more simultaneously visible AOIs, greater overall engagement, or both (as body visibility, AOI count, and AOI size covary here and cannot be separated). The divergent skin-visibility pattern (one participant increasing, one decreasing) may reflect individual differences in

Table 2: Hit rate by AOI and metric (undirected/free-viewing gaze task, n=2).

Stratification	Metric	P1	P2	Pooled
Overall				
All	Raw Gaze	0.111	0.080	0.096
	Fixation	0.097	0.059	0.077
	dwell-episode	0.096	0.063	0.079
By AOI				
Jawline/Neck	Raw Gaze	0.189	0.111	0.150
	Fixation	0.175	0.068	0.119
	dwell-episode	0.172	0.082	0.126
Clavicles	Raw Gaze	0.083	0.058	0.070
	Fixation	0.067	0.039	0.052
	dwell-episode	0.066	0.039	0.052
Upper Arms	Raw Gaze	0.066	0.058	0.062
	Fixation	0.052	0.032	0.041
	dwell-episode	0.055	0.048	0.051
Stomach/Waist	Raw Gaze	0.119	0.106	0.112
	Fixation	0.125	0.093	0.109
	dwell-episode	0.101	0.089	0.095
Hips	Raw Gaze	0.055	0.062	0.059
	Fixation	0.043	0.057	0.051
	dwell-episode	0.042	0.051	0.046
Thighs	Raw Gaze	0.111	0.068	0.090
	Fixation	0.069	0.067	0.068
	dwell-episode	0.097	0.057	0.077

Note. Raw gaze and dwell-episode hit rates share a denominator (all in-window gaze samples), so dwell-episode ≤ raw gaze. Fixation hit rate is computed over detected fixations. P1 and P2 are the two participants. Pooled combines both ($n = 2$). Movement-speed and skin-visibility breakdowns are summarized in-text.

response to skin exposure. Finally, the observation that dwell retention peaked during fast movement for jawline/neck, upper arms, and stomach/abdomen/waist, but not thighs, may indicate this is not simply a generic property of high-attention AOIs. A larger, BID-screened sample would clarify to what extent AOI-specific factors (e.g., body region, typical motion) contribute.

5.5 Limitations

This feasibility test used a convenience sample of seven women, split into validation groups of four (no validation), two (passed), and one (warned), which is too small for anything other than descriptive case observations. Because these rates pool eligible observations across participants, they are weighted toward participants who contributed more eligible observations and do not account for dependence among observations from the same participant. We therefore read them as descriptive aggregates, not inferential estimates. Four participants completed no calibration validation, since this step was introduced mid-way through testing. For the directed task, participant compliance was assumed but not verified, so tracker/calibration error could not be distinguished from non-compliance. In the full study, we will incorporate a directed-task compliance check, independent of the gaze-containment outcome, so that apparent tracker or calibration error can be better distinguished from failure to follow the gaze instruction. Bounding-box annotations exhibited frame-to-frame jitter, including large artificial centroid displacements at scroll transitions and rapid cuts. These artifacts impacted our exploratory pursuit analyses by making frame-level estimates of AOI motion unreliable and fragmenting AOI-relative fixations. Movement speed, body visibility, and skin visibility were naturalistic, time-varying video properties rather than independently manipulated factors, and were likely correlated with each other and with AOI salience as body visibility (full vs. upper) was confounded with AOI identity. With no control condition, patterns such as the jawline/neck prominence cannot be separated from generic free-viewing biases (e.g., screen-center or face/head bias). We also leveraged a consumer-grade Tobii Eye Tracker 5 whose sampling rate is lower than most research-grade eye trackers. Finally, all comparisons came from a single 3-minute auto-scrolling feed and may not generalize to other body-oriented SFV content with different framing, pacing, or body types.

6 Conclusion

This feasibility test suggests that, once a calibration-validation step is introduced, our pipeline may produce interpretable estimates of gaze on moving AOIs, even from a small sample using a low-cost, consumer-grade eye tracker. The resulting hit rates varied descriptively with calibration-validation status, AOI size, and movement condition. On this basis, we plan to proceed to a full study comparing the gaze patterns of women with and without BID-driven conditions, contingent on three refinements: (1) mandatory calibration validation for all participants, with recalibration until a passing result; (2) continued use of the dwell-episode hit rate as our primary metric for sustained gaze on moving AOIs for the ≈33 Hz gaze-point stream used in this study; and (3) a revised AOI annotation approach that reduces frame-to-frame bounding-box jitter and improves dwell-episode precision. Beyond these procedural refinements, the exploratory patterns we observed warrant testing in a larger, BID-screened sample beyond our hypothesis-generating n=2. Finally, our exploratory analyses did not provide a sufficiently stable or interpretable measure of smooth pursuit; however, because AOIs are extended regions that gaze can hold through saccadic sampling, sustained attention does not necessarily involve smooth pursuit, so distinguishing the precise oculomotor mechanism may not be necessary for measuring sustained gaze on moving AOIs. A higher-sampling-rate tracker and a revised annotation approach could nonetheless make the velocity-based metrics feasible in future work.

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